# Mer Bleue Ecosystem Evaluation Technical Report

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In fulfillment of EVS4010: EVS Field Course

Submitted to: Dr. Kyra St. Pierre, Brent Holmes

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University of Ottawa

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## 1) Introduction:

### 1.1 Wetland and Forest Ecosystems:

Wetlands are crucial ecosystems for environmental stability and human activity, accounting for 40.6% of the total global ecosystem services value (Xu and others 2020). The major services provided by wetlands are within the provisioning and regulating categories, including the provision of unique habitats for a variety of flora and fauna, and the regulation of atmospheric carbon (i.e. carbon sequestration), water quality and levels (i.e. filtration of polluted waters, flood mitigation), and groundwater (i.e. recharging of groundwater aquifers) (Mitsch and others 2015). Despite the importance of these ecosystems to biodiversity and human health, their conservation is greatly neglected, with more than 70% of global wetland areas deemed as "destroyed" or "impaired" due to anthropogenic causes (Kingsford and others 2016). These damages are mainly related to the change and expansion of land usages, such as urbanization and agricultural expansion, as well as the impacts of increased greenhouse effect due to human emissions (i.e. heatwave events, severe droughts) (Martinuzzi and others 2014).

Forest ecosystems are an additional crucial ecosystem, offering numerous ecosystem services and benefits. These benefits include climate regulation, carbon sequestration, support for biodiversity, improving water quality, and enabling elemental cycles such as the water and carbon cycles (Steel and others 2024).

### 1.2 Context of Study:

For the ecology module of EVS4010, we are focused on the biological observation and assessment of the Mer Bleue Bog, located within the Ottawa Green Belt. This area is maintained and protected by the National Capital Commission (NCC) and is recognized as a conservation area. Mer Bleue is a temperate ombrotrophic bog, meaning that it receives its water and nutrients solely from precipitation, and in the rise of current climate change impacts, its health is at a significant risk (Billett and others 2008). With this being said, we believe that the presence of the

beavers (*Castor canadensis*) in the area is crucial to the preservation and rehabilitation of Mer Bleue. The North American beaver is considered a keystone species regarding North American wetlands and offer a key indication of ecosystem health (Fischer and others 2025).

### 1.3 Objectives:

There are two main objectives that the following study is designed to answer. The first main objective is to identify the different types of underlying freshwater and forest ecosystems present at the Mer Bleue area. The second main objective is to answer the team's big picture question; Can the beaver (*Castor canadensis*) be an ally to the preservation of the Mer Bleue ecosystem in the future? Why or why not? This question will be answered alongside with reporting the many different ecosystems present to help guide and facilitate the remainder of the report. The study is designed to answer many sub-objectives. Some of which being, statistical analysis of the biodiversity present in the area, determination of the overall health of these ecosystems.

## 2) Methods:

### 2.1 Study Site Description:

Mer Bleue, recognized as a conservation area, is managed and protected by the NCC and was visited on August 21$^{st}$ and 22$^{nd}$. The approximate length of the trail is 1.5km and the bog itself spans 28 km$^2$ (Billett and Moore 2008). During the exploration around the site, it was determined that the parking lot and the pavilion area, at the start of the trail and the surrounding forests, were notable high points, while the bog and the marsh areas had the lowest elevation of the area. It was also noted that there were many trees within the forested areas and were absent around the main marsh area. After walking over the main marsh area, some smaller trees were present in the southern half of the trail after the main landing over the boardwalk. It was noticed that water was present in the main marsh area and was absent after reaching the main landing. Minimal water was present at the purple site, located in the bog area of the trail, however its

majority was present in the underlying peat. Water was absent in the forested areas, but this may be due to the extremely dry conditions in the Ottawa region over the summer. It is also important to mention that the area is rich in wildlife and vegetation, as there was a great presence of fauna such as chipmunks, birds, turtles, snakes, and even beavers scattered throughout.

The wetland study site of interest for the pink team was located on the far side of the main boardwalk, which was a part of the main marsh area. The coordinates for the location of the site are 45.391971°N 75.513286°W. On day one, there was nearby water flow in the open water occurring in the western direction. The percent of open water in the nearby surrounding area was 100%. The shape of the main marsh area can be viewed as a skinny oval. The main marsh area was bound on either side by a hill side, forest region, and by another arid bog region. There was a lot of evidence of animals and vegetation in the nearby area to the site, as there was a presence of turtles, frogs, birds, etc. There were no disturbances present in the area.

In the forested area of interest, it was located near the pavilion with the coordinates of 45.393204°N 75.513850°W. There was no water present in the area, and it seemed extremely arid as there was a lot of dead plant matter scattered throughout the site. It was full of vegetation on one side of the bridge and more barren on the other side. There was a presence of trees of various sizes from 3-15m in height. The canopy cover was underwhelming, as approximately 20% of the area was covered by vegetation above. This led to the site being extremely hot and sunny. The soil was damp, but not muddy. There was a great presence of wildlife such as insects were seen flying around and rodents roamed around the wooded area.

**2.2 Map of Study Area:**

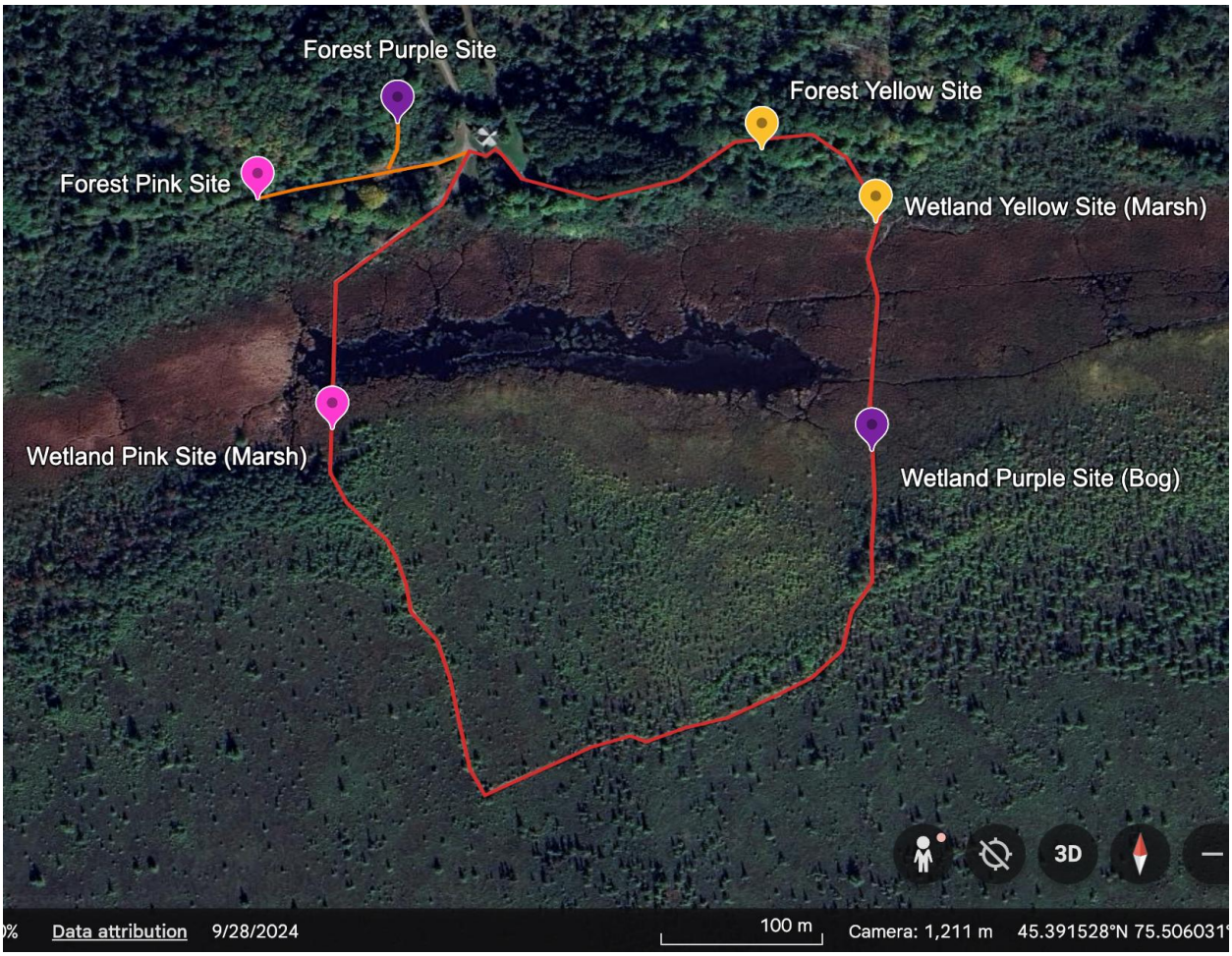


**Figure 1.** *Annotated Map of the Mer Bleue Study Area.* The information present in the map above being the respective wetland sites and forest sites. Along with the paths taken from the pavilion to these respective sites. Where the orange path shows the trails to the pink and purple forest sites. While the red path represents the path around the marsh and bog areas of Mer Bleue.

**2.3 Sampling Methods:**

On day 1, our field work included the conduction of biodiversity surveys (BioBlitz). The Bioblitz consisted of photographing the biodiversity at different sites around Mer Bleue and uploading these images to iNaturalist. iNaturalist is an application that utilizes artificial intelligence (AI) to identify the species' type of our images with moderate to high confidence.

Each team was assigned two different sites in order to evaluate the Mer Bleue ecosystem. With the use of a Quadrat (1x1m area) at five different locations, where the relevé method was

used to estimate the cover of each plant species. For each plant present in the quadrats the abundance, cover, sociability, vitality, fertility and phenology were considered during the surveying process at each respective site. Animals and trees were also noted during this process.Insect surveys were conducted through the usage of a net and dragged in and around the quadrats and the net was examined for insects. If present a photograph of the insect was taken and identified.

Day two in the field was designed for the collection and testing of water samples, along with characterization of soil profiles at respective sites. At each site a site overview, wetland hydrology, soil characteristics, and physical and chemical characteristics were completed. For the site overview and hydrology pictures and 360 videos were taken to help determine the parameters of each site of interest and to aid in the study description.

To evaluate soil characteristics, a soil core was taken, through the use of a soil corer, from the forest sites to determine the soil type, profile, and drainage classes. For the wetland sites, soil composition/type and drainage were also examined.

The physical chemical characteristics were determined using handheld pH and EC probe. Where the pH, conductivity, total dissolved solids, temperature. An oxygen meter was used for the determination of the % oxygen saturation at a given temperature. Three water samples were collected from the field using a bucket and stored in sampling bottles. These bottles were collected for further examination and testing in the lab. Three water samples were collected from the field using a bucket and stored in sampling bottles. These bottles were collected for further analysis in the lab.

**2.4 Treatment of Data:**

To evaluate the statistical differences in physiochemical parameters between wetland sites, a one-way ANOVA was conducted for each parameter (i.e. pH, DO, alkalinity) in RStudio. To evaluate alpha diversity for each site of the forest and wetland ecosystems, a Shannon Diversity Index value (H) was calculated in Excel. To compare the (dis)similarity between forest and wetland sites, a Bray – Curtis dissimarlity value (BC) was calculated in Excel. All data analysis, exclusive of statistical analysis, was conducted in Microsoft Excel.

### 2.5 Lab Methods:

The water samples collected at Mer Bleue on Day 2 were analyzed using various chemical instruments and techniques, including fluorometry, spectrophotometry, titration, colorimetry, and microscopy. To measure chlorophyll-a concentration, a fluorometer, set at 440 nm, was used to incur fluorescence. A spectrophotometer was used to measure the absorbance of the water sample in relation to its dissolved organic matter content. A colorimeter (Hach) was used to determine concentrations of Fe (II), Total Fe, $NO_3^-$, and $SO_4^{2-}$ . Titrations with 0.1 N HCl were performed to measure the alkalinity of the water samples. Finally, microscopes were used in the lab to identify plants collected in the field and to confirm the identity of the soil samples.

## 3) Results

### 3.1 BioBlitz Survey Results

**A. Wetland Sites**

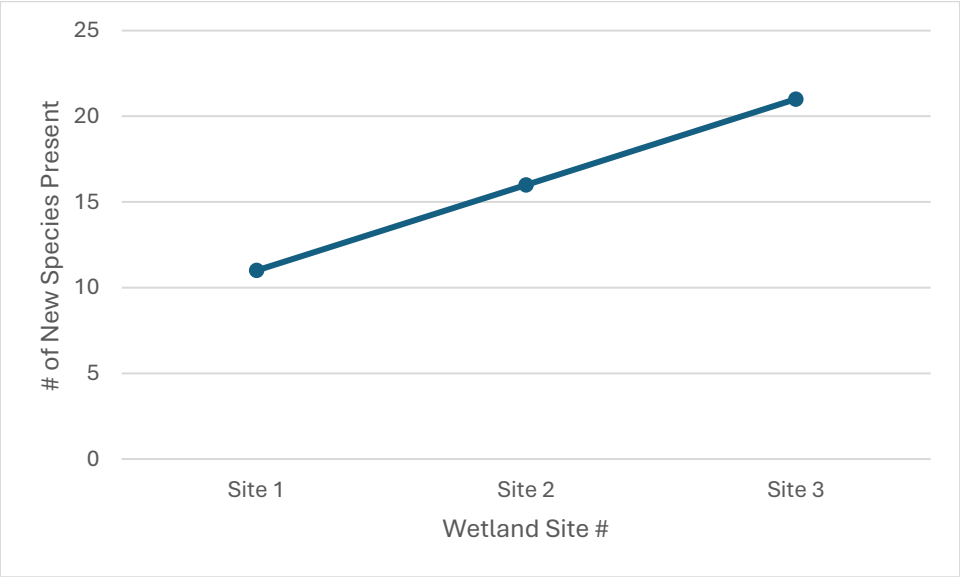


**Figure 2.** Species accumulation curve for 3 wetland sites at Mer Bleue Bog, Ottawa, ON. The y-axis presents a cumulative value for species present at each site (number of new species observed is added to the number of previously established species).

At Site 1 (Pink), we observed 11 species present, which acts as the basis of new species observed. At Site 2 (Purple), there were 5 new species present, resulting in 16 total species

present between the two sites. At Site 3 (Yellow), there were 5 new species present, resulting in 21 total species present between the three sites.

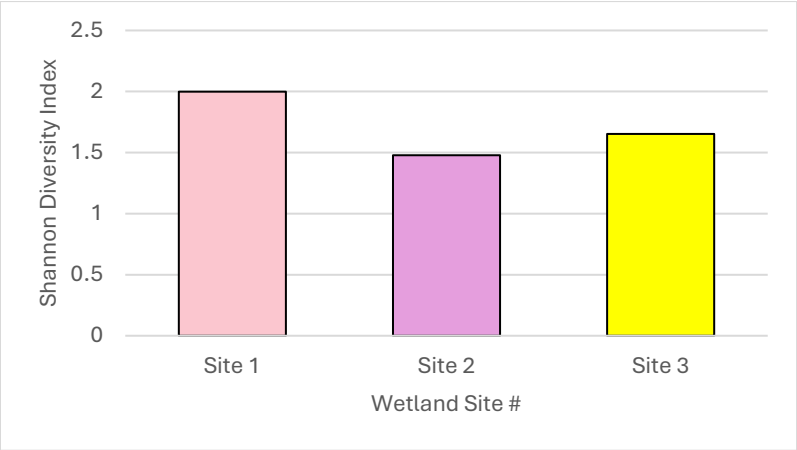


**Figure 3.** Shannon Diversity Index (alpha diversity metric) calculated for each wetland site at Mer Bleue Bog, Ottawa, ON.

To evaluate alpha diversity, we calculated the Shannon Diversity Index values (H) based on the mean species proportion at each site, for a total of 3 sites. For Site 1, 2, and 3, the H values calculated are 2.0, 1.5, and 1.7, respectively.

**B. Forest Sites**

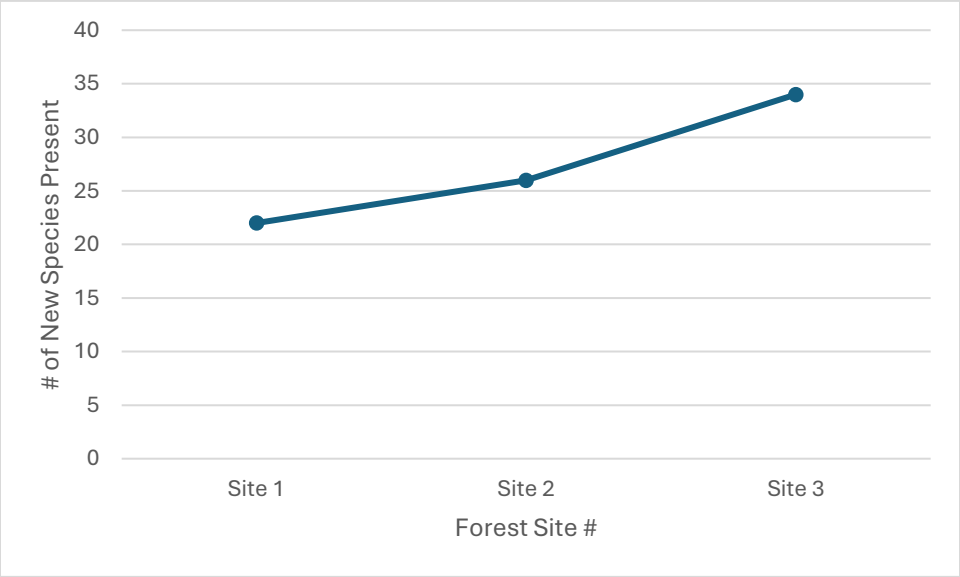

**Figure 4.** Species accumulation curve for 3 forest sites at Mer Bleue Bog, Ottawa, ON. The y-axis presents a cumulative value for species present at each site (number of new species observed is added to the number of previously established species).

At Site 1 (Pink), we observed 22 species, acting as the basis for the new species present. At Site 2 (Purple), there was 4 new species present, resulting in a total of 26 species present between the two sites. At Site 3 (Yellow), there was 8 new species present, resulting in a total of 34 species observed between the three sites.

**Figure 5.** Shannon diversity index (alpha diversity metric) calculated for each forest site at Mer Bleue Bog, Ottawa, ON

To evaluate alpha diversity, we calculated the Shannon Diversity Index values (H) based on the mean species proportion at each site, for a total of 3 sites. For Site 1, 2, and 3, the H values calculated are 4.5, 1.8, 2.4, respectively.

## C. Comparison Between Wetland and Forest

To evaluate the beta diversity between our wetland and forest sites, we calculated the Bray – Curtis dissimilarity value to be 0.83 (BCD = 0.83), indicating a low occurrence of shared species.

### 3.2 Physiochemical Data – Wetland Sites

**Table 1.** Water sampling parameters measured at 3 wetland sites (A, B, and C) at Mer Bleue Bog, Ottawa, ON. Each value, exclusive of water depth, represents the averaged value of replicate measurements for each parameter and the standard deviation (SD) of these replicates.

| Wetland Site | Mean Temp. ± SD (°C) | Mean pH ± SD | Mean Conductivity ± SD (µs/cm) | Mean TDS* ± SD (ppm) | Mean Dissolved Oxygen ± SD (%sat) | Water Depth (cm) |
|---|---|---|---|---|---|---|
| **A** | 18.26 ± 0.41 | 6.45 ± 0.085 | 166.78 ± 1.87 | 83.00 ± 0.54 | 32.30 ± 6.40 | 16 |
| **B** | 23.59 ± 0.21 | 5.71 ± 0.014 | 96.00 ± 1.79 | 48.00 ± 0.58 | 35.90 ± 4.10 | 70 |
| **C** | 18.77 ± 0.27 | 4.23 ± 0.042 | 120.0 ± 2.30 | 59.22 ± 0.96 | 6.56 ± 2.04 | 36** |

*TDS = Total Dissolved Solids

** Surface water depth was minimal (<1 cm); however, water depth measured through peat is represented by this value

Three wetland sites (A, B, and C) were chosen to measure water quality parameters, including water temperature, pH, conductivity, total dissolved solids (TDS), dissolved oxygen (DO) percent solubility, and water depth (refer to Table 1). Water sampling was conducted periodically throughout the morning, starting at 9:45 AM and completed by 11:45 AM, for all three sites. At Site A, we observed the highest pH (6.45), conductivity (166.78 µs/cm), and TDS (83 ppm) between all sites, and the lowest values for water temperature (18.26°C) and water depth (16 cm) between all sites. At Site B, we observed the highest values for water temperature (23.59°C), dissolved oxygen (35.9% sat), and water depth (70 cm), and the lowest values for conductivity (96 µs/cm) and TDS (48 ppm). At Site C, we observed the lowest values for pH (4.23) and dissolved oxygen (6.56% sat) between all sites. Through statistical analysis, it was shown that there are significant differences for pH ($F = 16.54$, $p = 0.00362$), and DO ($F = 7.273$, $p = 0.0249$) between the 3 sites, highlighting the difference between ecosystem types.

**Table 2.** Biological and chemical analysis of water samples collected at 3 wetland sites (A, B, and C) at Mer Bleue Bog, Ottawa, ON. The standard deviation of replicate measurements is represented as SD.

| Wetland Site | Mean Alkalinity ± SD (mg $CaCO_3$/L) | DOC* (abs) | Chlor-a (RFU) | [Fe (II)] (mg/L) | [Total Fe] (mg/L) |
|---|---|---|---|---|---|
| **A** | 41.03 ± 2.86 | 1.29 | 6435.61 | 0.39 | 3.78 |
| **B** | 1.85 ± 0.025 | 0.415 | 3171.36 | 0.08 | 1.42 |
| **C** | 12.25 ± 14.17 | 0.846 | 32775.5 | 0.88 | 2.8 |

*DOC represents dissolved organic carbon

Water samples were retrieved from each wetland site and analyzed for various biological (dissolved organic carbon (DOC), and chlorophyll-a fluorescence) and chemical parameters (alkalinity, ferrous iron (Fe (II)) concentration, and total iron concentration) to assess water quality. Site A showed the highest values for alkalinity (41.03 ± 2.86 mg CaCO3/L), DOC (1.29), and total Fe concentration (3.78 mg/L) between all sites. Site B showed the lowest values for alkalinity (1.85 ± 0.025 mg CaCO3/L), Fe (II) concentration (0.08 mg/L), and total Fe concentration (1.42 mg/L) between all sites. Site C showed the highest values for DOC (0.846), chlorophyll-a fluorescence (32775.5 RFU), and Fe (II) concentration (0.88 mg/L). Through statistical analysis, there is a significant difference in alkalinity between the 3 sites ($F = 17.75$, $p = 0.00302$). Additional chemical parameters were measured, including concentrations of sulfate and nitrate, but were absent in all sample analyses (below level of detection).

### 3.3 Physiochemical Data – Forest Sites

Table 3. Forest ecosystem characteristics assessed at 3 sites at Mer Bleue Bog, Ottawa, ON.

| Forest Site | Soil Type | Average Canopy Cover (%) | Soil Drainage Type |
|---|---|---|---|
| **A** | Sandy clay loam | 80 | High |
| **B** | Silty clay loam | 15 | Poor |
| **C** | Sandy loam | 80 | Moderate |

At each forest site (A, B, and C), we identified the main type of soil present, the level of soil drainage, and the average canopy cover. Site A was composed mainly of sandy clay loam, representing high soil drainage, and an average canopy cover of 80%. Site B was composed mainly of silty clay loam, representing poor soil drainage, and an average canopy cover of 15%. Site C was composed mainly of sandy loam, representing moderate soil drainage, and an average canopy cover of 80%. With these results, Site B shows the lowest soil drainage level and the lowest canopy cover, while Site A and C share fairly similar characteristics.

## 4) Discussion:

### 4.1 Primary Vegetation in Mer Bleue

The plant species prevalent among the pink wetland site were notably the narrow-leaf cattails (*Typha angustifolia*), sensitive ferns (*Onoclea sensibilis*), and leather leaf's (*Chamaedaphne calyculata*). The sensitive fern tends to thrive among shaded, damp, and high organic matter soil,

(Anderson 2022) and the narrow-leaf cattails were observed among all riparian intermediates. Lastly, the leather leaf characteristically grows among peat bogs, where there is limited nutrient availability, and tolerate low pH soil (Źróbek-Sokolnik and others 2019). These observations are indicative of marsh-to-bog-like areas.

The three prevalent species observed within the yellow wetland site were the leatherleaf (*Chamaedaphne calyculata*) and labrador tea (*Rhododendron neoglandulosum*). These observations were expected, as these sites are located among marsh and bog-like areas. As mentioned previously, the leather leaf resides within peat bogs. Labrador tea is also found to predominantly grows in bogs within the boreal forests in Canada, favouring nutrient-poor, low pH soils (Hébert and Thiffault 2011).

Lastly, the most prominent vegetation observed among the purple wetland site was sphagnum moss (*Sphagnum sp.*). Sphagnum moss was predominantly observed in this site due to it's proximity to the bog. Sphagnum moss is also referred to as Bog Moss, which predominantly grow among bogs and in wetlands where low pH and sandy soil characteristics and moist environments. A unique physiological adaptation that sphagnum moss possesses is that as the organism grows, the top region continues to grow and while the rest will decompose, accumulating into peat (Missouri Department of Conservation 2024). Low dissolved oxygen levels were yielded at the purple wetland site. This would be representative of anoxic conditions within the water, when microbes resort to using anaerobic respiration. This was additionally indicative by a subtle sulfur odour among the site.

Observations made at the forest sites showed a difference from its wetland counterpart at Mer Bleue. A high number of vegetation and wildlife was observed compared to the wetland sites. It is important however, to note that the forest and wetland species found in the pink sites did share four common species among each other, which was represented by a low beta diversity value. Different tree species were identified at all three sites. The glossy buckthorn (*Frangula alnus*), black cherry (*Prunus serotina*), and choke cherry (*Prunus virginia*) were identified among the pink forest site. Red maple (*Acer rubrum*), Meadowsweets (*Spiraea sp.*), and Lindens (*Tilia sp*.) were observed within the purple forest site. Gray alder (*Alnus incana*), and False Buckthorn (*Frangula alnus*) were additionally identified among the yellow forest site. Additionally, soil

moisture was found to be highest among the pink forested site. This was reflective of the proximity to the wetlands, where water had evidently been previously and can be explained by the higher biodiversity. The difference in vegetation diversity found in the forest sites and the wetland sites represents the contrast in the conditions required for certain vegetation and wildlife to adapt and reside in.

**4.2 Statistical Analysis of Species Diversity**

In our evaluation of forest species diversity through the Shannon Diversity Index, we determined high species diversity in Site 1 (Pink), moderate diversity in Site 3 (Yellow), and low diversity in Site 2 (Purple). We observed 22 different species of flora and fauna in Site 1, supporting its high diversity index, 9 species in Site 2, with only 5 new species present, and 7 species in Site 3, with also 5 new species present. This stark contrast in diversity between sites is primarily due to the moisture level in the soil and soil type, as Site 1 had moderately moist soil with poor drainage, whereas Sites 2 and 3 had more arid soils with moderate to high drainage. Moreover, Sites 2 and 3 had a high canopy cover (80%) compared to Site 1 (15%), creating more specific growth requirements for vegetation (i.e. low exposure to sunlight).

In our evaluation of wetland species diversity, we determined all wetland sites to be of relatively high species richness, but low species diversity, according to the calculated Shannon Diversity indices. From the 3 observed wetland sites, Site 1 (Pink) showed the highest diversity, with 11 different species present, and Site 2 (Purple) with the lowest diversity, with only 7 species present. This is in accordance with our observations in the field, as Site 1 was located adjacent to open water and Site 2 was completely dried-up. This difference in water presence plays a major role in wetland species diversity and the ability of species present in Site 1 to flourish in Site 2.

Based on these observations of the prevalent species in the wetland and forest sites, the identification of the different ecosystems observed at Mer Bleue are marsh and bog areas surrounded by a boreal forest.

### 4.3 The Importance of Beavers in the Ecosystem at Mer Bleue

The beaver (*Castor canadensis)* plays a paramount role in wetland and riparian ecosystems. As a keystone species in riparian ecosystems (Fischer and others 2025), beavers possess a unique ability to promote growth in biodiversity, sustaining wetland habitats and strengthening surrounding waterways (Parks Canada Agency 2019). The modern north American beaver transforms and maintains their surrounding habitat, being categorized as an allogenic ecosystem engineer, which characteristically alters their habitats, inadvertently strengthening the surrounding ecosystem (Fairfax and Westbrook 2024). The beavers' adaptations and evolutionary behaviours have led to building dams, cutting down surrounding vegetation such as riparian hard woods, and burrowing holes in the soil. Additionally, in areas where beavers cut down trees to use for dams and scrounge for food, the riparian hardwoods were discovered adapting to regrowing quickly, sprouting soon after being chewed off.

In the context of the beaver's role at Mer Bleue, a drought was evident during the period of observation. Beavers contribute to the preservation of wetland ecosystems especially during periods of dryness. From the infrastructure that is built, dams produce beaver ponds which provide the capacity to reserve and increase water storage capacity and its residence time (Fairfax and Westbrook 2024). Dams also improve the water quality of wetlands by reducing erosion and maintaining the essential sediments that aid in filtering heavy metals and other anthropogenic activities (Parks Canada Agency 2019). However, because the Mer Bleue beaver pond system is relatively smaller in size compared to other wetland areas in North America, the surrounding areas affected by the increase in temperature and drought had further dried. Due to the lack of precipitation this summer, Mer Bleue's bog had appeared to dry significantly. Without the aid of the beaver population within Mer Bleue, the biodiversity would have further collapsed, as the water storage would have dwindled.

### 4.4 Limitations of the Study

Limitations during the period of study occurred during the collection of samples and laboratory testing. During the collection of samples, the area of interest had been unusually dry due to lack of precipitation compared to previous years. This made collecting water samples at certain sites especially difficult. Additionally in previous years, there had been higher biodiversity observed,

however because this was a particularly dry year, many organisms were not identified at the sites such as fungus which require an increase in moisture content availability to thrive. Samples collected from the field were kept in coolers overnight which may have altered certain parameter values that were tested and analyzed. During laboratory analysis, the phosphate colorimetry tests was not able to be tested due to the absence of the testing material.

**5) Conclusion:**

Over the past few days after conducting biodiversity surveys, soil characteristics, physical/chemical characteristics, and analysis of water samples in the lab of various sites scattered throughout the Mer Bleue. It was determined that there are three different types of ecosystems present in the area. These ecosystems being a marsh, a bog, and a boreal forest. It was also determined that there are many different types of the soils present in the area such as sandy clay loam with a high drainage, silty clay loam with a low drainage, and sandy loam with moderate drainage. After the completion of the statistical analysis, it was also determined that there was a high species diversity at the pink site, a moderate diversity at the yellow site, and a low diversity at the purple site. The evaluation of the wetland species diversity using the Shannon diversity indices, determined that all the wetland sites had a relatively high species diversity. Regarding the big picture question given to the team. It was established that the beavers are crucial to the preservation of the Mer Bleue area. It is unfortunate that the study period took place after an extremely dry summer which lead to a major limitation during the study as the sampling may have not been representative of what should have been seen during this time period.

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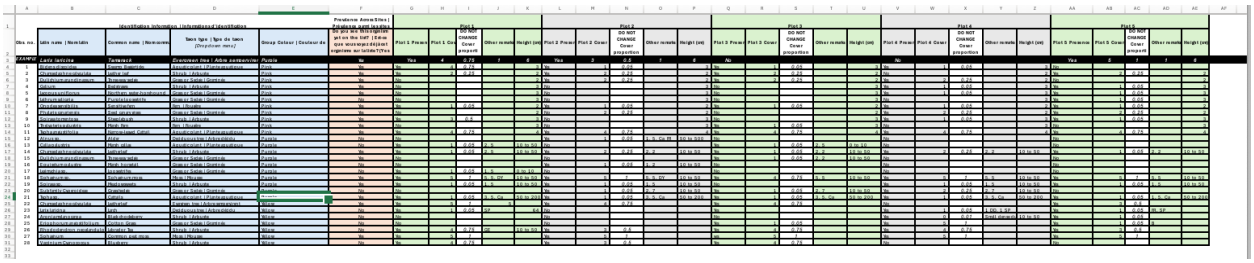
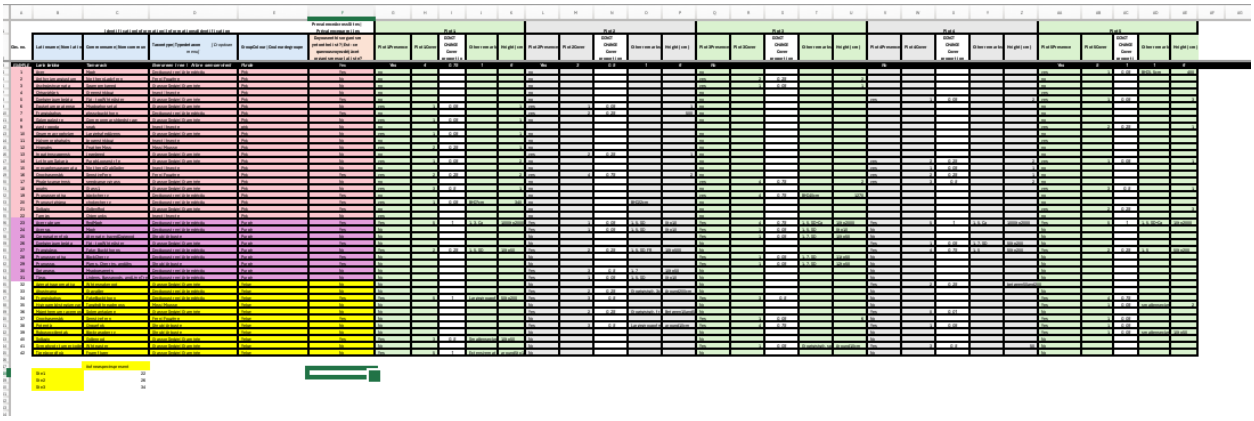
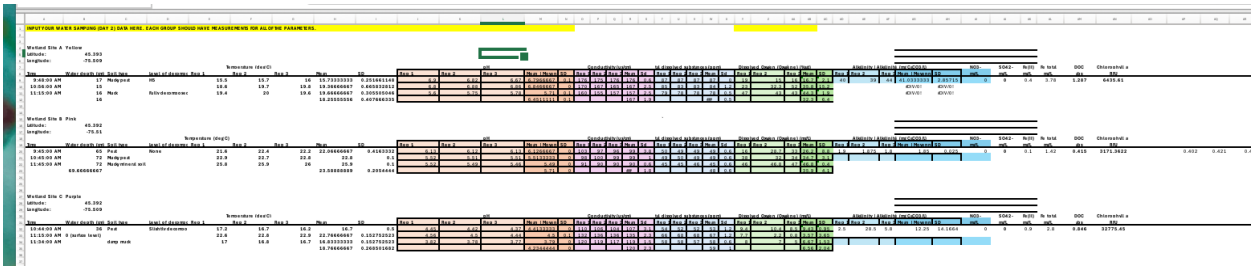
# Appendix:







| Site ID | Latitude | Longitude | Soil type \| Type de sol | Average Canopy Cover at Site \| Couverture moyenne au site | Other notes \| Autres notes |
|---|---|---|---|---|---|
| Yellow | 45.3933 | -75.51 | Sandy clay loam | 80 | Top layer is darker, bottom layer, starting at 13.2cm depth is lighter in colour. Bottom layer is more dense and stiff. The soil is weell drained. |
| Pink | 45.393204 | -75.51385 | Silty Clay Loam | 15 | Poor drainage, no water present but soil was damp/moist |
| purple | 45.39361111 | -75.51277778 | Sandy loam | 0.8 | Soil drainage class: somewhat excessive Soil remains in ball when squeezed, does not shine, < 1 inch ribbon, very gritty texture |

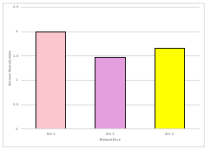
DO NOT CHANGE. THESE ARE LINKED TO COLUMNS IN THE BIODIVERSITY TABS AND CAN BE USED FOR THE CALCULATION OF ALPHA AND BETA DIVERSITY METRICS.

## WETLAND | ZONE HUMIDE

PROPORTIONS (ABUNDANCE PROXY)

| Latin name | Common name | Taxon type | Site ID | Plot 1 | Plot 2 | Plot 3 | Plot 4 | Plot 5 | Mean Proportion at Site | Standardized | pi*ln(pi) | pi^2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bidens dis | Swamp Beggarticks | Aquatic plant \| Plante aquatique | Pink | 0.75 | 0.05 | 0.05 | 0.05 | | 0.23 | 0.10 | -0.2344 | 0.050625 |
| Chamaeda | Leather leaf | Shrub \| Arbuste | Pink | 0.25 | 0.25 | 0.25 | | 0.25 | 0.25 | 0.11 | -0.2484 | 0.062500 |
| Dulichium | Three-way sedge | Grass or Sedge \| Graminée | Pink | | 0.25 | 0.25 | 0.25 | | 0.25 | 0.11 | -0.2484 | 0.062500 |
| Galium | Bedstraws | Shrub \| Arbuste | Pink | | | | 0.05 | 0.05 | 0.05 | 0.02 | -0.0866 | 0.002500 |
| Lycopus us | Northern water-horehoun | Grass or Sedge \| Graminée | Pink | | | | 0.05 | 0.05 | 0.05 | 0.02 | -0.0866 | 0.002500 |
| Lythrum sa | Purple loosestrife | Grass or Sedge \| Graminée | Pink | | | | | 0.05 | 0.05 | 0.02 | -0.0866 | 0.002500 |
| Onoclea se | Sensitive fern | Fern \| Fougère | Pink | 0.05 | 0.05 | 0.05 | 0.05 | 0.05 | 0.05 | 0.02 | -0.0866 | 0.002500 |
| Phalaris ca | Seed canary grass | Grass or Sedge \| Graminée | Pink | | 0.25 | | 0.25 | 0.25 | 0.25 | 0.11 | -0.2484 | 0.062500 |
| Spiraea ton | Steeplebush | Shrub \| Arbuste | Pink | 0.5 | | | 0.05 | 0.05 | 0.20 | 0.09 | -0.2192 | 0.040000 |
| Thelypteris | Marsh Fern | Fern \| Fougère | Pink | | | 0.05 | | | 0.05 | 0.02 | -0.0866 | 0.002500 |
| Typha anal | Narrow-leaved Cattail | Aquatic plant \| Plante aquatique | Pink | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 | 0.34 | -0.3671 | 0.562500 |
| Alnus sp. | Alder | Deciduous tree \| Arbre décidu | Purple | | 0.05 | | | 0.05 | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Calla palus | Marsh callas | Aquatic plant \| Plante aquatique | Purple | 0.05 | | 0.05 | | 0.05 | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Chamaeda | Leatherleaf | Shrub \| Arbuste | Purple | 0.05 | 0.25 | 0.05 | 0.25 | 0.05 | 0.13 | 0.08 | -0.2079 | 0.016900 |
| Dulichium | Three-way sedge | Grass or Sedge \| Graminée | Purple | | | 0.05 | | | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Equisetum | Marsh horsetail | Grass or Sedge \| Graminée | Purple | | 0.05 | | | | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Lysimachu | Loosestrifes | Grass or Sedge \| Graminée | Purple | 0.05 | | | | | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Sphagnum | Sphagnum moss | Moss \| Mousse | Purple | 1 | 1 | 0.75 | 1 | 1 | 0.95 | 0.61 | -0.3 | 0.902500 |
| Spiraea sp. | Meadowsweets | Shrub \| Arbuste | Purple | 0.05 | 0.05 | | 0.05 | 0.05 | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Subtamiv r | Grass/sedge | Grass or Sedge \| Graminée | Purple | | 0.05 | 0.05 | 0.25 | | 0.12 | 0.08 | -0.1947 | 0.013611 |
| Typha sp. | Cattails | Aquatic plant \| Plante aquatique | Purple | 0.05 | 0.05 | 0.05 | 0.05 | 0.05 | 0.05 | 0.03 | -0.1108 | 0.002500 |
| Chamaeda | Leatherleaf | Evergreen tree \| Arbre sempervivent | Yellow | 1 | 0.75 | 0.75 | | 0.5 | 0.75 | 0.21 | -0.3296 | 0.562500 |
| Larix laricin | Larch | Deciduous tree \| Arbre décidu | Yellow | 0.05 | | | 0.05 | 0.05 | 0.05 | 0.01 | -0.0606 | 0.002500 |
| Aronia melo | Black chokeberry | Shrub \| Arbuste | Yellow | | | | 0.01 | 0.05 | 0.03 | 0.01 | -0.0407 | 0.000900 |
| Eriophorum | Cotton Grass | Grass or Sedge \| Graminée | Yellow | | | 0.05 | 1 | 0.05 | 0.37 | 0.10 | -0.236 | 0.136444 |
| Rhododend | Labrador Tea | Shrub \| Arbuste | Yellow | 0.75 | 0.5 | 0.75 | 0.75 | 0.5 | 0.65 | 0.19 | -0.3123 | 0.422500 |
| Sphagnum | Common peat moss | Moss \| Mousse | Yellow | 1 | 1 | 1 | 1 | 1 | 1.00 | 0.28 | -0.3577 | 1.000000 |
| Vaccinium | Blueberry | Shrub \| Arbuste | Yellow | 0.75 | 0.5 | 0.75 | | | 0.67 | 0.19 | -0.3155 | 0.444444 |

| | Shannon Weaver Index | Simpson Index |
|---|---|---|
| Site 1 | 1.9996 | 1.172361172 |
| Site 2 | 1.478 | 1.052065555 |
| Site 3 | 1.6525 | 0.389519961 |



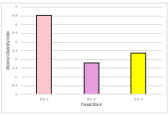
| Wetland Common name | Wetland P | standard wetland P | Forest P | standard forest p | In Forest | ABS DIFF | SUM | BCD |
|---|---|---|---|---|---|---|---|---|
| Swamp Beggarticks | 0.23 | 0.10 | 0 | 0 | no | 0.10 | 0.103211 | 0.827617597 |
| Leather leaf | 0.25 | 0.11 | 0 | 0 | no | 0.11 | 0.114679 | |
| Three-way sedge | 0.25 | 0.11 | 0 | 0 | no | 0.11 | 0.114679 | |
| Bedstraws | 0.05 | 0.02 | 0.05 | 0.011609928 | yes | 0.01 | 0.034537 | |
| Northern water-horehoun | 0.05 | 0.02 | 0 | 0 | no | 0.02 | 0.022936 | |
| Purple loosestrife | 0.05 | 0.02 | 0.12 | 0.027068832 | yes | 0.00 | 0.050005 | |
| Sensitive fern | 0.05 | 0.02 | 0.42 | 0.096674401 | yes | 0.07 | 0.139610 | |
| Seed canary grass | 0.25 | 0.11 | 0.63 | 0.145011601 | yes | 0.03 | 0.259691 | |
| Steeplebush | 0.20 | 0.09 | 0 | 0 | no | 0.09 | 0.091743 | |
| Marsh Fern | 0.05 | 0.02 | 0 | 0 | no | 0.02 | 0.022936 | |
| Narrow-leaved Cattail | 0.75 | 0.34 | 0 | 0 | no | 0.34 | 0.344037 | |
| Maple | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Northern Lady fern | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |
| Swamp milkweed | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Flat-top White Aster | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Meadow horsetail | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| glossy buckthorn | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |
| snails | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |
| Lage leafed Avens | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Feather Moss | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |
| jewel weed | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |
| Northern Crab Spider | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Grass 1 | 0 | 0 | 0.50 | 0.116009281 | | 0.12 | 0.116009 | |
| black cherry | 0 | 0 | 0.75 | 0.174013921 | | 0.17 | 0.174014 | |
| choke cherry | 0 | 0 | 0.05 | 0.011609928 | | 0.01 | 0.011601 | |
| Golden Rod | 0 | 0 | 0.25 | 0.05800464 | | 0.06 | 0.058005 | |

## FOREST | FORÊT

PROPORTIONS (ABUNDANCE PROXY)

| Latin name | Common name | Taxon type | Site ID | Plot 1 | Plot 2 | Plot 3 | Plot 4 | Plot 5 | Mean Proportion at Site | pi*ln(pi) |
|---|---|---|---|---|---|---|---|---|---|---|
| Acer | Maple | Deciduous tree \| Arbre décidu | Pink | | | | | 0.05 | 0.05 | -0.1498 |
| Athyrium angustum | Northern Lady fern | Fern \| Fougère | Pink | | | 0.25 | | | 0.25 | -0.3466 |
| Asclepias incarnata | Swamp milkweed | Grass or Sedge \| Graminée | Pink | | | 0.05 | | | 0.05 | -0.1498 |
| Chinavia hilaris | Green stink bug | Insect \| Insecte | Pink | | | | | | #DIV/0! | |
| Doellingeria umbellata | Flat-top White Aster | Grass or Sedge \| Graminée | Pink | | | | 0.05 | 0.05 | 0.05 | -0.1498 |
| Equisetum pratense | Meadow horsetail | Grass or Sedge \| Graminée | Pink | 0.05 | 0.05 | | | | 0.05 | -0.1498 |
| Frangula alnus | glossy buckthorn | Deciduous tree \| Arbre décidu | Pink | | 0.25 | | | | 0.25 | -0.3466 |
| Galium palustre | Common marsh bed | Grass or Sedge \| Graminée | Pink | 0.05 | | | | | 0.05 | -0.1498 |
| gastropoda | snails | Insect \| Insecte | pink | | | | | 0.25 | 0.25 | -0.3466 |
| Geum macrophyllum | Large leafed Avens | Grass or Sedge \| Graminée | Pink | 0.05 | | | | | 0.05 | -0.1498 |
| Halyomorpha halys | brown stink bug | Insect \| Insecte | Pink | | | | | | #DIV/0! | |
| Hypnales | Feather Moss | Moss \| Mousse | Pink | 0.25 | | | | | 0.25 | -0.3466 |
| Impatiens capensis | Jewel weed | Grass or Sedge \| Graminée | Pink | | 0.25 | | | | 0.25 | -0.3466 |
| Lythrum Salicaria | Purple Loosestrife | Grass or Sedge \| Graminée | Pink | 0.05 | | | 0.25 | 0.05 | 0.12 | -0.2507 |
| mecaphesa asperata | Northern Crab Spider | Insect \| Insecte | Pink | | | | 0.05 | | 0.05 | -0.1498 |
| Onoclea sensibilis | Sensitive Fern | Fern \| Fougère | Pink | 0.25 | 0.75 | | 0.25 | | 0.42 | -0.3644 |
| Phalaris canariensis | seed canary grass | Grass or Sedge \| Graminée | Pink | | | 0.75 | 0.5 | | 0.63 | -0.2938 |
| poales | Grass 1 | Grass or Sedge \| Graminée | Pink | 0.5 | | | | 0.5 | 0.50 | -0.3466 |
| Prunus serotina | black cherry | Deciduous tree \| Arbre décidu | Pink | | | 0.75 | | | 0.75 | |
| Prunus virginiana | choke cherry | Deciduous tree \| Arbre décidu | Pink | 0.05 | | | | | 0.05 | -0.1498 |
| Solidago | Golden Rod | Grass or Sedge \| Graminée | Pink | | | | | 0.25 | 0.25 | -0.3466 |
| Tamias | Chipmunks | Insect \| Insecte | Pink | | | | | | #DIV/0! | |
| Acer rubrum | Red Maple | Deciduous tree \| Arbre décidu | Purple | 1 | 0.05 | 0.75 | 1 | 1 | 0.76 | -0.2086 |
| Acer sp. | Maple | Deciduous tree \| Arbre décidu | Purple | | 0.05 | 0.05 | | | 0.05 | -0.1498 |
| Cornus alternifolia | Alternate-leaved Dog | Shrub \| Arbuste | Purple | | | 0.05 | | | 0.05 | -0.1498 |
| Doellingeria umbellata | Flat-top White Aster | Grass or Sedge \| Graminée | Purple | | | | 0.05 | | 0.05 | -0.1498 |
| Frangula sp. | False-Buckthorns | Deciduous tree \| Arbre décidu | Purple | 0.25 | 0.25 | | 0.75 | 0.25 | 0.38 | -0.3673 |
| Prunus serotina | Black Cherry | Deciduous tree \| Arbre décidu | Purple | | | 0.05 | | | 0.05 | -0.1498 |
| Prunus sp. | Plums, Cherries, and | Shrub \| Arbuste | Purple | | | 0.05 | | | 0.05 | -0.1498 |
| Spiraea sp. | Meadowsweets | Deciduous tree \| Arbre décidu | Purple | | 0.5 | | | | 0.50 | -0.3466 |
| Tilia sp. | Lindens, Basswoods | Deciduous tree \| Arbre décidu | Purple | | 0.05 | | | | 0.05 | -0.1498 |
| Agastrina aromatica | White snakeroot | Grass or Sedge \| Graminée | Yellow | | | | 0.25 | | 0.25 | -0.3466 |
| Alnus incana | Gray alder | Deciduous tree \| Arbre décidu | Yellow | | 0.25 | | | | 0.25 | -0.3466 |
| Frangula alnus | False Buckthorn | Deciduous tree \| Arbre décidu | Yellow | 1 | 0.5 | 0.5 | | 0.75 | 0.69 | -0.2576 |
| Hypnoamblystedium var | Tangled thread moss | Moss \| Mousse | Yellow | | | | | 0.05 | 0.05 | -0.1498 |
| Maianthemum racemos | Solomon's plume | Grass or Sedge \| Graminée | Yellow | | 0.25 | | 0.01 | | 0.13 | -0.2652 |
| Onoclea sensibilis | Sensitive fern | Fern \| Fougère | Yellow | | | 0.05 | | 0.05 | 0.05 | -0.1498 |
| Potentilla | Cinquefoils | Shrub \| Arbuste | Yellow | | 0.5 | 0.75 | 0.05 | 0.05 | 0.34 | -0.3666 |
| Rubus occidentalis | Black raspberry | Shrub \| Arbuste | Yellow | | | | | 0.05 | 0.05 | -0.1498 |
| Solidago | Goldenrod | Grass or Sedge \| Graminée | Yellow | 0.5 | | | | | | |
| Symphyotrichum ericoides | White aster | Grass or Sedge \| Graminée | Yellow | | | 0.05 | 0.5 | | 0.28 | -0.355 |
| Tiarella cordifolia | Foam flower | Grass or Sedge \| Graminée | Yellow | 1 | | | | | 1.00 | 0 |

| | Shannon Weaver Index |
|---|---|
| Site 1 | 4.535489632 |
| Site 2 | 1.821676245 |
| Site 3 | 2.386947173 |

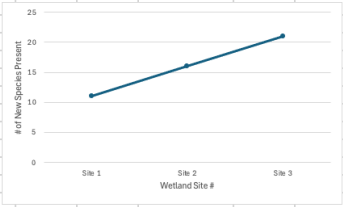


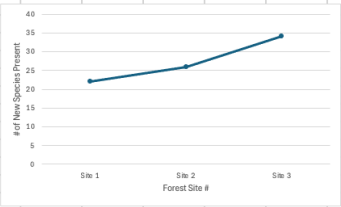
TIONS (ABUNDANCE PROXY)

| ot 3 | Plot 4 | Plot 5 | Mean Proportion at Site |
|---|---|---|---|
| | | 1 | 0.75 |
| 0.05 | 0.05 | | 0.23 |
| 0.25 | | 0.25 | 0.25 |
| 0.25 | 0.25 | | 0.25 |
| | 0.05 | 0.05 | 0.05 |
| | 0.05 | 0.05 | 0.05 |
| | | 0.05 | 0.05 |
| 0.05 | 0.05 | 0.05 | 0.05 |
| | 0.25 | 0.25 | 0.25 |
| | 0.05 | 0.05 | 0.20 |
| 0.05 | | | 0.05 |
| 0.75 | 0.75 | 0.75 | 0.75 |
| | | | 0.05 |
| 0.05 | | | 0.05 |
| 0.05 | 0.25 | 0.05 | 0.13 |
| 0.05 | | | 0.05 |
| | | | 0.05 |
| | | | 0.05 |
| 0.75 | 1 | 1 | 0.95 |
| | 0.05 | 0.05 | 0.05 |
| 0.05 | 0.25 | | 0.12 |
| 0.05 | 0.05 | 0.05 | 0.05 |
| 0.75 | | 0.5 | 0.75 |
| | 0.05 | 0.05 | 0.05 |
| | 0.01 | 0.05 | 0.03 |
| 0.05 | 1 | 0.05 | 0.37 |
| 0.75 | 0.75 | 0.5 | 0.65 |
| 1 | 1 | 1 | 1.00 |

| Site# | # of new species | |
|---|---|---|
| Site1 | 11 | 11 |
| Site2 | 5 | 16 |
| Site3 | 5 | 21 |

| | # of new species present |
|---|---|
| Site1 | 22 |
| Site2 | 26 |
| Site3 | 34 |




started with pink site (most species present)

started with pink site (most species present)

Alpha Diversity

| site | temp | temp_sd | pH | pH_sd | conduct | conduct_sd | TDS | TDS_sd | DO | DO_sd | Alk |
|---|---|---|---|---|---|---|---|---|---|---|---|
| yellow | 15.7333333 | 0.25166115 | 6.79666667 | 0.11676187 | 175.666667 | 0.57735027 | 87 | 0 | 16.6666667 | 2.081666 | 39.8 |
| yellow | 19.3666667 | 0.66583281 | 6.84666667 | 0.04163332 | 167.333333 | 2.51661148 | 83.6666667 | 1.15470054 | 35.8 | 15.1561869 | 39 |
| yellow | 19.6666667 | 0.30550505 | 5.71 | 0.09643651 | 157.333333 | 2.51661148 | 78.3333333 | 0.47140452 | 44.3 | 1.91572441 | 44.3 |
| pink | 22.0666667 | 0.4163332 | 6.12666667 | 0.0057735 | 98.6666667 | 3.7859389 | 49.3333333 | 0.57735027 | 26.1666667 | 8.7785724 | 1.85 |
| pink | 22.8 | 0.1 | 5.51333333 | 0.0057735 | 99 | 1 | 49.3333333 | 0.57735027 | 34.6666667 | 3.05505046 | 1.875 |
| pink | 25.9 | 0.1 | 5.49 | 0.03 | 90.3333333 | 0.57735027 | 45.3333333 | 0.57735027 | 46.7666667 | 0.35118846 | 1.825 |
| purple | 16.7 | 0.5 | 4.41333333 | 0.04041452 | 106.666667 | 3.05505046 | 52.6666667 | 1.15470054 | 9.43333333 | 0.9504385 | 2.5 |
| purple | 22.7666667 | 0.15275252 | 4.5 | 0.06 | 134.666667 | 2.30940108 | 67.3333333 | 1.15470054 | 3.56666667 | 3.64737348 | 28.5 |
| purple | 16.8333333 | 0.15275252 | 3.79 | 0.02645751 | 118.666667 | 1.52752523 | 57.6666667 | 0.57735027 | 6.66666667 | 1.52752523 | 5.75 |